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Lost-In-Space: Star Pattern Recognition and Attitude Estimation for the Case of No Prior Attitude Information

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Jer-Nan Juang^{*}, and Daniele Mortari[⊕]

A novel attitude determination approach is presented and results from night sky validation tests are discussed. The central ideas are extensions of the *K-Vector* method recently introduced by Mortari. It involves a construction of a judicious “star pair catalog” prior to launch, wherein all cataloged stars are considered as star pairs and are ordered $\{k = 1, 2, \dots, N \sim O(10^6)\}$, over the whole sky, sorted in the order of increasing inter-star angle. From this, we have a “searchless” means to access the candidate set of stars for each measured star pair, but this *K-Vector* method may still give 10s and sometimes 100s of candidate stars for each measured pair. We introduce in this paper a method to identify the measured stars in the subset of candidate stars accessed using the *K-vector*. The new method is based upon a logical process which *pivots* about two stars (the first identified pair) to more efficiently identify or ignore the remaining measured stars. Using this Pivot Method, we show that star identification can be reliably and efficiently implemented on-orbit, even for the Lost-In-Space case, and thereby this paper introduces a globally valid process, consistent with real-time, on-board computational constraints, that solves the most fundamental problem associated with star pattern identification. Results from night sky experiments are discussed which support the validity of the analysis and practicality of this approach. Also discussed are plans for on-orbit experiments. The StarNav experiment is planned for Shuttle Mission STS 107, January 2000; this will represent the first ever on-orbit demonstration of a star sensor implementing a Lost-In-Space star identification and attitude determination process.

Introduction

Star observations are widely used by spacecraft as a primary means of precision attitude determination. Star line of sight vectors are usually measured by a CCD (Charge Coupled Device) camera generating a 2D image of a small section of the sky. Since the first star tracker based on the use of CCD area array imaging sensors was developed in the early 1970s at JPL (Jet Propulsion Laboratory), many families of solid state star sensors have been developed to meet the desire for increasingly precise, real-time, on-board attitude estimation during the last two decades. Currently, CCD based star trackers allow us to obtain precise spacecraft attitude estimates by applying star pattern recognition algorithms in an on-board computer, which has been implemented with several existing algorithms.

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However, attitude control, pointing control, and navigation systems of future advanced spacecraft will be characterized by requirements for a high degree of autonomy, automatic on-orbit calibration, very high accuracy, efficient commandability, and fast fault recovery. These characteristics are incompatible with the constraints of conventional star trackers and current flight operational algorithms which require at least approximate prior attitude estimates or, in many cases, additional approximate sensors (such as sun, horizon, or magnetic field sensors) and associated algorithms. In addition to being inadequate for implementing ambitious plans for future spacecraft systems, conventional star trackers and algorithms are costly to use because their lack of flexibility/adaptability necessitates the expenditure of significant human, ground software, uplink, redundant sensors, and telemetry resources.

With the advent of more accurate and sensitive CCDs, faster microprocessors, and higher-density memory chips, it has become feasible to build an *autonomous, intelligent* star tracker and software system which is capable of alleviating the above-mentioned problems of existing systems. Accompanying the development of star tracker and microprocessor hardware, many different approaches for star identification and associated algorithms have been devised to solve the attitude determination problem. Nevertheless, only a few star pattern identification methods, especially for the *lost-in-space* case, have been developed. No algorithm capable of solving the lost-in-space problem has been implemented for an actual space mission. Thus the present algorithm, scheduled to be tested on-orbit aboard STS-107 in early 2001, will represent a significant milestone.

Van Bezooijen [5] developed an autonomous star identification method for the AST (Autonomous Star Tracker) developed and tested by Lockheed Palo Alto Research Laboratory (Currently, Lockheed Martin's Advanced Technology Center). The algorithm, which was also designed to consider the lost-in-space case, finds the largest group of observed stars that matches a group of guide stars and attempts to maximize its success rate by exploiting the full information content of star patterns including the visual magnitude information. The guide star database was specially designed, which consists of the position and brightness of the acquisition guide stars plus a sorted list of all acquisition guide star pairs that fit in the FOV (Field-Of-View) of AST.

Mortari [11] developed a robust star identification approach, called the SLA (Search-Less Algorithm), for the *lost-in-space* case, which solves a problem common to all star identification techniques. By introducing the $\mathbf{K}$ -vector technique, Mortari's method [11] eliminates the star catalog searching phase which represents the heaviest computational load of existing star identification procedures. This has enabled new *whole sky* star pattern recognition approaches that *do not require either prior sub-catalogs, or any searching phase to find candidate stars for pattern matching*. The $\mathbf{K}$ -vector method is a general procedure that directly addresses a global data base that corresponds to, for example, $\sim 10^5$ stars in a mission catalog. The $\mathbf{K}$ -vector method can be adapted for almost any existing approach and improve it dramatically in terms of speed. The method enables real-time non-iterative solution of the problem: Identify the indices of all the data x_i satisfying a given requirement $x_i \in [x_{\min}, x_{\max}]$, within any n -long data vector $\mathbf{x}$. The original SLA approach

consists of two different identification processes: **K**-vector star-pair identification and *Reference-Star* matching identification. The **K**-vector star-pair identification process, which is common and essential to the **K**-vector based star identification methods, selects an admissible star-pair set from the star-pair catalog whereas the subsequent reference star matching identification process performs the final identification sequence by evaluating angular separation combinations with a subset of all of the admissible star-pairs. In order to find the reference-star, the identification process of this approach mainly relies on two major indices of the highest occurrences by computing a histogram of the overall admissible index vectors. Once the reference star pair is identified, it is possible to proceed with the identification of the other observed stars by the analysis of the index vectors corresponding to the reference vector [11]. In order to increase the efficiency in the identification process for multiple FOV star cameras, the strategy of the SLA has been modified [14], a method that finds common indices among subsets of measured star triads instead of using the reference star as in [11].

Mortari and Junkins [14] introduced the SP-Search (Spherical Polygon-Search) approach to identify the stars as observed by a wide FOV star tracker, which uses more than once the **K**-vector technique, does not require any (accurate or not) initial guess of the spacecraft attitude, does not use magnitude information, and is particularly suitable for the *lost-in-space* general case. The method is based upon simultaneously matching interstar angles for three stars; first the two vectors for a reference observed star pair are adopted to create a vector basis with which the star identification process is accomplished, by projecting the vectors locating for all of the other remaining ($n-2$) observed stars, and using the measurement uncertainty to identify the usually unique third star (of any measured triad) in the **K**-vector accessed subset of candidate stars. In the unique formulation of [14], this is accomplished in a searchless fashion, however some rules-based logic is required to operate on the **K**-vector accessed stars to make the algorithm essentially failsafe. The performance of this method is maximized when interstar angles approach 90° , so it is attractive for multiple FOVs star trackers.

While the **K**-Vector method gives direct searchless access to a candidate set of stars for each measured star pair, this method may still provide 10s up to 100s of candidate stars from the master mission catalog, for each measured pair, and thus a significant number of stars must still be considered in the final star identification logic. In this paper, we introduce a method to deal with multiple measured pairs, an idea based upon *pivot measured stars*. Using the new *Pivot Method*, we show that star identification can be reliably implemented on-orbit, for the *Lost-In-Space* case, and thereby we establish a globally valid process, consistent with real-time, on-board computational constraints, that solves the most fundamental problem associated with star pattern identification. The whole sky star identification procedure is explained in the following sections, which includes the generation of the star pair catalog using K-vector technique, **K**-vector star-pair identification, and pivot method based star identification.

Star Pair Catalog Generation using K-Vector Technique

The **K**-vector technique has been devised in order to avoid the searching phase characterizing essentially all existing star pair identification algorithms. This vector directly provides the indices of the admissible star-pairs range, as discussed briefly below and in detail in [11]. The method is general, and can be applied to any sorted n -long real

vector $\mathbf{Y}$, in which we want to identify a subset of elements of $\mathbf{Y}$, specifically, all of the elements falling between $(\nu-\delta)$ and $(\nu+\delta)$. It involves a construction of a judicious “star pair catalog” prior to launch, wherein all catalogued stars are considered as star pairs are ordered $\{k = 1, 2, \dots, N \sim O(10^6)\}$, over the whole sky, indexed continuously in the order of increasing inter-star angle. Mortari [11] has shown that, given a measured interstar angle and an associated maximum star centroiding error, then a deterministic, non-iterative access using the K -Vector information gives the k indices of the feasible (*all possible*) star pairs from the catalog, and therefore, the master catalog indices using $(i = I(k), j = J(k))$ of all possible cataloged star pairs from the whole sky, that could have the measured interstar angle. Thus if the measured star pair is in the catalog, we can be certain that it is contained in the non-iteratively accessed subset of stars.

The master star catalog adopted for this study is the catalog used for the Near Earth Asteroid Rendezvous Mission (NEAR); this NEAR catalog was constructed from the Smithsonian Astrophysical Observatory (SAO) bright star catalog by T. E. Strikwerda et al at the Johns Hopkins University Applied Physics Laboratory (JHU/APL). This catalog contains 12,620 stars covering the magnitude range from -1.5 up to 6.4 with a magnitude precision of ~ 0.2 in which each star has stored a unique integer name, its visual magnitude, and three components of direction cosines. The catalog is considered to have worst case errors of a small fraction of an arc second in star positions. This precision can be considered perfect for most practical purposes of star identification, since state of the art star cameras have about one order of magnitude larger errors associated with image centroiding. In our case, we have single star centroiding errors that average about 5 arc seconds, one sigma, and image stars down to about visual magnitude 6.5. Thus the NEAR catalog is well-matched with our sensor precision and sensitivity. Figure 1 shows a histogram of the number of the 12,620 stars in the NEAR master catalog vs. visual magnitude.

Construction of the K -vector star *pair catalog* can be initiated by pre-calculating the interstar sine values of all combinations of catalogued stars within the given cut-off threshold of visual magnitude. For convenience, we use the square of the interstar sine values to avoid calculation with the trigonometric functions as well as to easily use components of direction cosines as given in the master star catalog.

$$\begin{aligned} \mu_{ij} &= \sin^2 \Theta_{ij} = |\hat{\mathbf{u}}_i \times \hat{\mathbf{u}}_j|^2 = (u_{yi}u_{zj} - u_{zi}u_{yj})^2 + (u_{zi}u_{xj} - u_{xi}u_{zj})^2 + (u_{xi}u_{yj} - u_{yi}u_{xj})^2 \\ P &= \{\dots \sin^2 \Theta_{ij} \dots\}^T, \quad I_P = \{\dots i \dots\}^T, \quad J_P = \{\dots j \dots\}^T \end{aligned} \quad (1)$$

P is a one dimensional array, sorted with increasing values of $\mu_{ij} = \sin^2 \Theta_{ij}$.

This pair catalog can be truncated, deleting pairs with interstar angle ranges that could not possibly be measured due to a particular star camera system’s field of view and similar geometric considerations. We typically restrict the pair catalog to ignore stars closer together than $\theta_{\min}$ and further apart than $\theta_{\max}$. For a single 10° circular field of view camera, clearly $\theta_{\max} = 10^\circ$ and usually we take $\theta_{\min} \cong 0.5^\circ$.

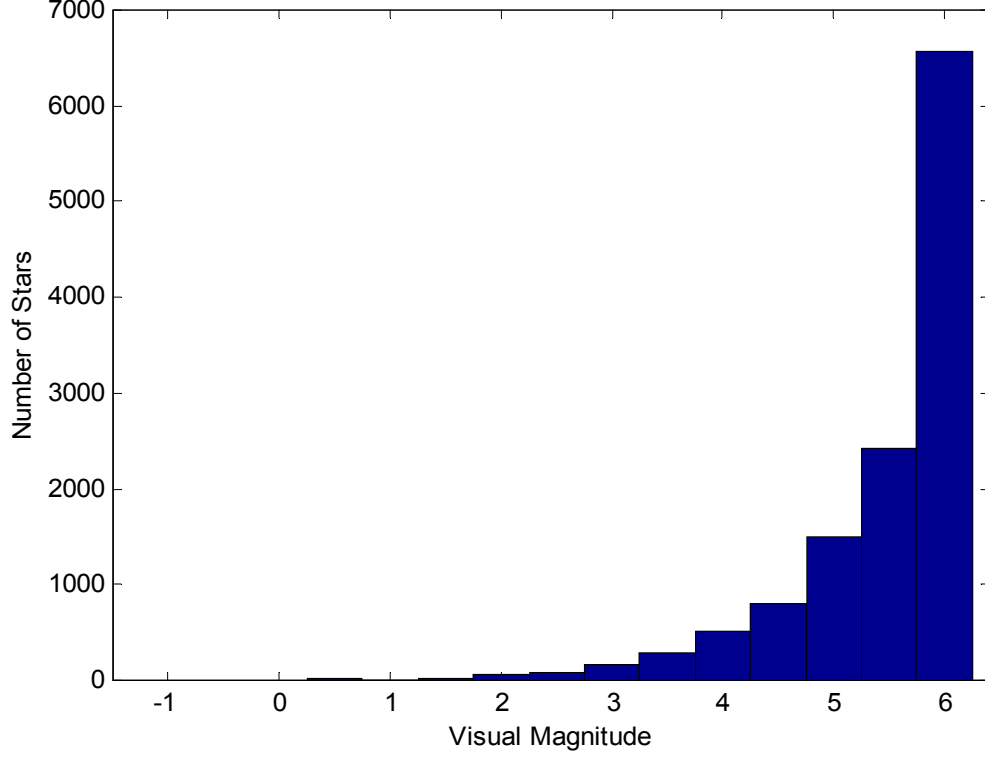


Figure 1 Star Distribution versus Visual Magnitude in the NEAR Master Catalog

Any star-pair that passes the logical truncation: $\sin^2 \theta_{\min} \leq |\hat{u}_i \times \hat{u}_j|^2 \leq \sin^2 \theta_{FOV} = \sin^2 \theta_{\max}$ is called “admissible” and is retained in the final P array. The size (N) of the resulting *star pair catalog* P is the total number of star pairs in the master catalog that could possibly be imaged by the camera, considering the full sky. Clearly N is mainly dominated by the size of FOV and the cut-off threshold value of visual magnitude (which is in turn dictated by the sensor integration time, CCD quantum efficiency, and lens design). The star pair catalog is constructed from the (say) 12,000 star catalog through a computationally expensive one-time search process over the entire catalog; the stars are ordered in increasing interstar angle (i.e., starpair number 1 will have a separation angle of about $\theta_{\min}$ and star pair number N will have a separation angle of about $\theta_{\max}$). Obviously, this star pair catalog is constructed in a one-time ground-based computation and then stored in the flight computer. With a typical setting of parameters, for our camera, we find $N \sim 10^6$. In order to avoid the cases such that the number of measured stars is not enough to perform the successful pattern identification, the star sensor FOV and integration time are selected carefully to assure a sufficient average number of observed stars. The prototype of the *StarNav* camera used for our current research has a 10 deg x 10 deg FOV and an integration times from 0.01 sec, the cut-off magnitude is about 6.5, extensive tests indicate these choices result in 5 or more of imaged stars in the FOV for virtually all pointing directions, and typically, we image about 20 stars. Figure 2 shows the average number of measured stars vs. FOV, the relationship between the size of star pair catalog and the cut-off threshold level of visual magnitude according to the FOV of star tracker.

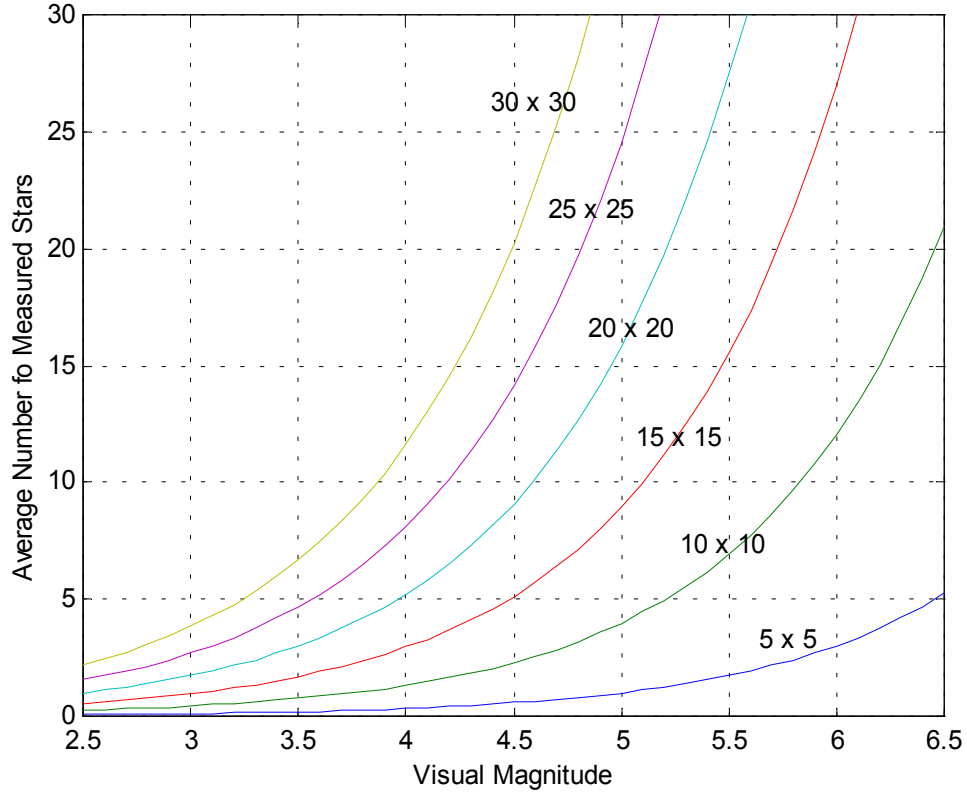


Figure 2 Average Number of Measured Stars within given FOV

When constructing the star pair catalog, in a pair of separate *index pointer arrays* (I, J), a pair of integers are stored for each (the k^{th}) admissible star pair in this sequence, giving the integer indices ($I(k) = i, J(k) = j$) of the $(i, j)^{\text{th}}$ stars in the master star catalog. That is, all the admissible star pairs, with which the associated integer indices maintain the corresponding relationships, are arranged in ascending order as

$$Y = \{\dots P(k) \dots P(l) \dots\}^T, \quad I = \{\dots I_p(k) \dots I_p(l) \dots\}^T, \quad J = \{\dots J_p(k) \dots J_p(l) \dots\}^T \quad (2)$$

So once k is known, we immediately have the master catalog indices of both stars in the pair from the index arrays $\{i = I(k), j = J(k)\}$. These arrays can be built as discussed below.

Following Mortari [11], we find the straight line connecting the two extremes $[1, Y(1)]$ and $[n, Y(n)]$, has, on average, only one element $Y(i)$ for each $D = [Y(n) - Y(1)]/(n-1)$ step. Let us consider the slightly steeper line which connects the two points $[1, Y(1) - D/2]$ and $[n, Y(n) + D/2]$. Because this straight line is found [11] to be an excellent approximation, it simplifies the process by avoiding many index checks and directly locates the desired elements of $Y(i)$. The equation of this line is

$$\sin^2 \Theta_{ij} = a_1 k + a_0, \quad \text{where} \quad \begin{cases} a_1 = nD/(n-1) \\ a_0 = Y(1) - a_1 - D/2 \end{cases} \quad (3)$$

where $i=1-n$. Starting with $K(1)=0$, an integer vector K is then built as follows

$$K(i) = j \quad \text{where} \quad Y(j) \leq a_1 i + a_0 \leq Y(j+1) \quad (4)$$

From a practical point of view, the i -th element of the K vector represents the number of elements $Y(i)$ below the value $Y(i)=a_1 i + a_0$. Once this vector is built, the evaluation of the two indices identifying, in the Y vector, the range of all the possible catalog star-pairs matching with the observed one, becomes an easy and fast task. The main details of the procedure to find the admissible range is explained in the following section. The specifications of the K -vector star pair catalog generated by this procedure are summarized in the Table 1.

Table 1 K-Vector Star Pair Catalog Specifications

	Circular FOV		Square FOV	
FOV size(deg)	10		10 x 10	
Visual Magnitude Threshold	5.5	6.0	5.5	6.0
Number of Stars	4787	8146	4787	8146
Dimension of K-vectors	186027	541132	369413	$\sim 10^6$

K-vector based Star-Pair Identification

The K -vector star-pair identification procedure starts by accessing the candidate set of cataloged stars for each measured star pair. For any observed (p, q) star-pair, a number of h admissible star-pairs, which can be arranged to form the h -dimensional index vector arrays $[I, J]$, are obtained from the star pair catalog. The dimension of h is to be determined by how many star pairs exist such that interstar sine angle of each star pair falls within the measured value, plus or minus the sensor accuracy threshold.

Once the angular separation and sensor accuracy are given for any measured star-pair, the indices associated with these values in the Y vector are simply provided as

$$\begin{cases} j_{bot} = bot\{[\sin^2(\theta - \delta) - a_0] / a_1\} \\ j_{top} = top\{[\sin^2(\theta + \delta) - a_0] / a_1\} \end{cases} \quad (5)$$

where the function $top\{z\}$ is defined as the larger integer number next to z , and $bot\{z\}$ is the smaller integer number immediately below z . Once j_{bot} and j_{top} are evaluated, it is possible to compute the bounding indices

$$k_{start} = K(j_{bot}) + 1 \quad \text{and} \quad k_{end} = K(j_{top}) \quad (6)$$

where k_{start} and k_{end} represent the extremes of the index set

$$k_{start} \leq k \leq k_{end} \quad (7)$$

This set contains the indices of all possible cataloged star pairs that have interstar angles $\theta \pm \gamma$. The $\mathbf{K}$ -vector can easily be applied to the star pairs identification. The $\mathbf{Y}$ vector, contains the sorted sine squared of all the observable star pair angles, and the $(\gamma - \delta)$ and $(\gamma + \delta)$ range limits represent $\sin^2 \theta \pm \delta$, where $\delta = \gamma \sin(2\theta)$, respectively, where θ is the angle between two observed stars and γ is the sensor precision (we adopt for γ the 4 - sigma centroiding error for an individual star, in radians). Note δ is the variation of $\sin^2 \theta$ due to a perturbation of θ by γ . The much smaller $\mathbf{K}$ -vector completely substitutes the vector $\mathbf{Y}$ in that it contains pointers to only the subset of star pairs that *could possibly* have been measured. Ref. [11] contains a more detailed presentation and discussion of the $\mathbf{K}$ -vector technique.

The size of the admissible range for the associated admissible star-pair set $[k_{\text{start}}, k_{\text{end}}]$ in Eq. (7) mainly depends on the precision of the star tracker and the prescribed visual magnitude threshold. According to Mortari's work the average value of the admissible star pair number, that is, the average value of $h (= k_{\text{start}} - k_{\text{end}} + 1)$ is less than 2 for a reference star tracker having $\text{FOV} = 30$ deg, $M_T = 3$, and the sensor centroiding precision of 10 arcsec. However, since the number of the admissible star-pairs $(k_{\text{end}} - k_{\text{start}} + 1)$ quickly increase for a more sensitive sensor (lower magnitude threshold), we find that when using a cutoff at visual magnitude 6 typically gives $(k_{\text{end}} - k_{\text{start}} + 1)$ on the order of 10s up to 100s of candidate star pairs for each measured pair. Thus it is evident that further judicious logic is required to finalize the star identification, given these easily accessed candidate sets of stars for each measured pair.

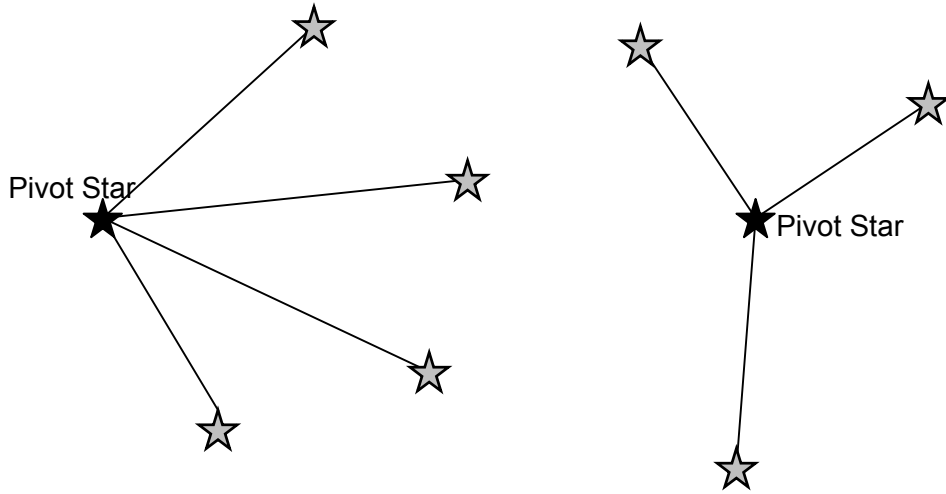


Figure 3 Schematic Diagram of the Pivot Method for Star Identification

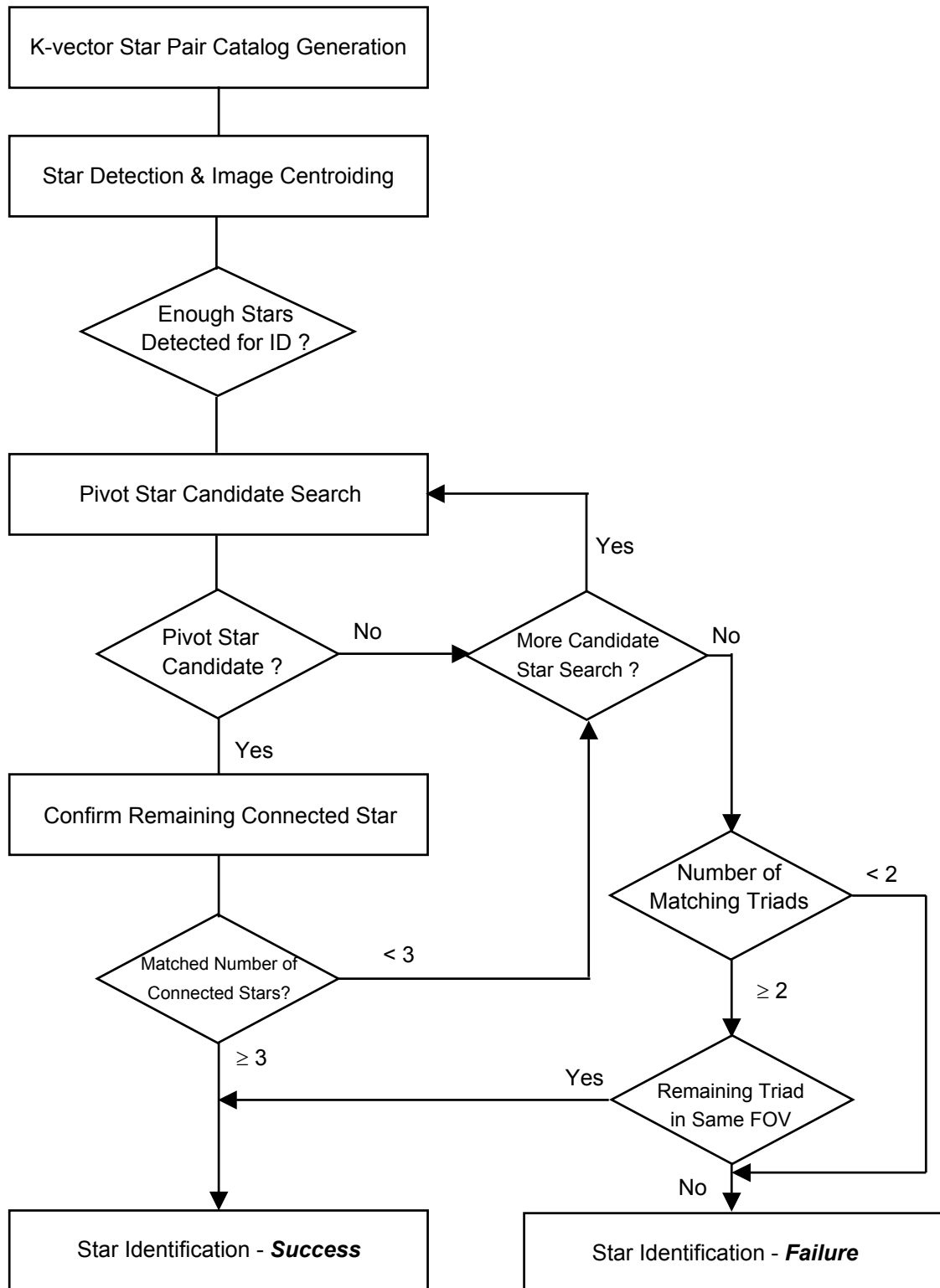
Pivot Method for Star Identification

The star pattern recognition approach used in this paper is mainly derived from the modified SLA (Search Less Algorithm) identification method. The main strategy of the modified SLA is as follows: given subsets of measured star triads, three different combinations can be used to obtain the admissible star pair set with $\mathbf{K}$ -vector method. Since the true index of each measured star must appear at least twice in the whole set of

admissible ranges, star identification can be considered as a success if we have the index set occurring twice in each set of admissible ranges. For sub-ten-arc second centroiding errors, this method has been found to fail occasionally and then only for the case of near double stars, and we have precluded this most likely failure mode by deleting all star pairs closer than θ_{min} from our star pair catalog. By having five or more measured stars, and insisting that all interstar angles between a minimum of four measured stars match the corresponding cataloged stars, we have found this scheme virtually never gives incorrect identifications. However, occasionally this conservative identification approach will fail to identify stars when there are only a small number of measured stars in the FOV. For high frame rate cameras, an occasional failure is generally acceptable, because use of a Kalman Filter to recursively operate on the attitude estimates will provide robustness with respect to occasional dropouts of the star-derived precision attitude estimates. Obviously, some index comparison logic over the small $\mathbf{K}$ -vectors subset of accessed stars is required to find the matching star indices from two or more $\mathbf{K}$ -vector accessed subsets of cataloged stars. The Pivot method is designed to accelerate this process by greatly increasing the probability to near-certainty that the same star is contained in the two subsets of star pair indices being compared, and upon location of the index match, we will immediately identify the additional measured stars paired with the pivot stars.

A flowchart of the pivot method for star identification is depicted in Fig. 4. First, it is checked if a sufficient number of stars (at least, 4) is available to enable successful identification. If a sufficient number of measured stars is available for identification, search for the pivot star candidate can be then initiated. To acquire the pivot star, a triad matching procedure from the modified SLA identification method is adopted. Brightest star among the first matching triad subset is normally chosen as a candidate of the pivot star. If even one subset of triad stars is not matched, the star identification is reported as a *failure*. Once the pivot star is chosen, it is possible to proceed with the identification of the additional (that is, the 3rd) measured star connected to the pivot star by checking the common indices among the admissible $\mathbf{K}$ -vector index set. If at least 3 stars are confirmed as connected to the pivot star, the star identification is considered to be *successful*. If there are more than 2 ambiguous set of triads, which are not identified by the additional match with the additionally connected stars, star identification can be confirmed by checking whether they are in the same FOV with their direction cosine information.

Figure 4 Flowchart of the Pivot Method for Star Identification



Results from Ground Validation Experiments

To evaluate further the validity and practicality of the pivot method, a family of night sky experiments were performed with the first prototype of StarNav micro star tracker. The first StarNav prototype is a low cost star tracker for micro spacecraft developed at Texas A&M University (TAMU) [15, 16], which has a 10° of single FOV with a magnitude 6.5 sensitivity and has shown excellent performance results from the night sky experiments recently. Its camera head unit has a 70 mm f/1.0 lens, 512 x 512 CCD detector with high quantum efficiency, and a thermoelectric cooler. The imager is an Apogee AP7 CCD camera (Apogee Instruments) which is interfaced to the computer via an ISA bus card that also carries the camera's power supply modules. The Octagon PC500 embeddable single board computer manufactured by Octagon Systems has a Pentium-133 processor with 32Mb RAM and a 40Mb "solid state disk" as well as conventional drives.

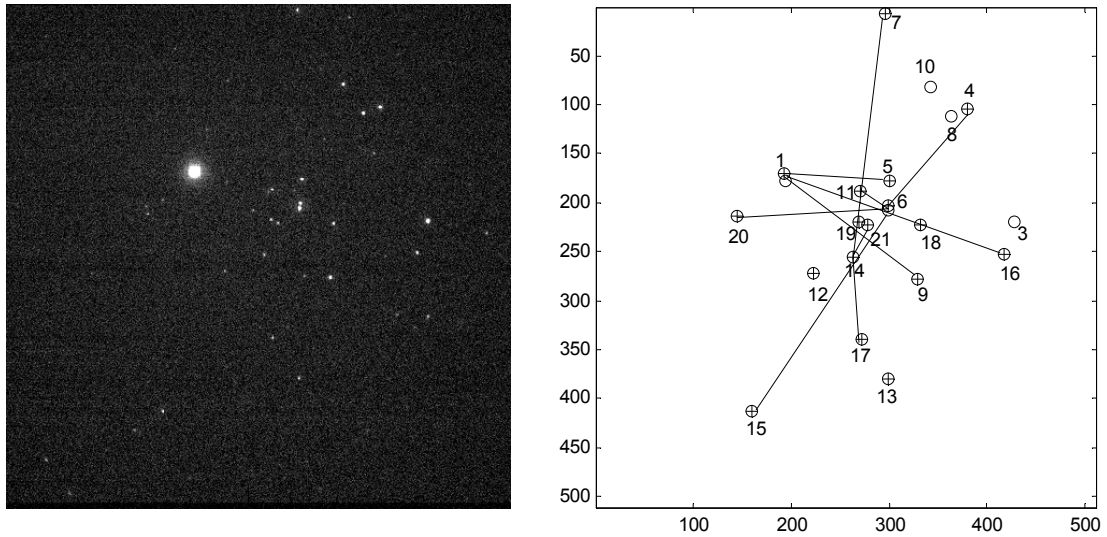
This 16-bit grey scale camera was selected primarily because it uses a thinned, back illuminated CCD sensor (SITE) which has an unusually high QE (0.85 peak and averaging >0.7 over the wavelength range $380 < \lambda < 900$ nm). Following laboratory validation and integration with appropriate optics, the camera was configured on an equatorial telescope mount, so it could be easily swept across the sky with known orientation; and the entire system validated end-to-end, under night sky viewing conditions.

Two night sky images were selected for validation of the pivot method, as regards the discussion in this paper. Also, the performance of and differences between star identification using the *pivot method* and the *modified SLA method* are discussed. Each image was taken with 10 ms exposure near Taurus and Lyra, respectively. The two methods were applied to the first star image (Fig. 5), in which include a very distinctively bright star 'Taurus'. The advantage of the triad (or triangle) match based on the modified SLA method is that triangle patterns are more distinct than pairs, thus eliminating many spurious stars, and handling triad groups is much more manageable. Even though the uniqueness probability for matching star triads is several orders of magnitude less likely to result in an invalid match than using a single star pair star pair, occasionally (less than a fraction of a percent of the time) the algorithm fails to correctly match the correct triangle and invokes a totally different catalog star triad in a incorrect sky location; this is more likely when the precision tolerance is too loosely prescribed. The subset of triads in which the 3 brightest stars are included result in an incorrect identification in the totally different location of the sky when the modified SLA was applied with a loose precision tolerance ($\gamma = 3\sigma = 0.003$ deg ~ 11 arc sec), whereas the star identification was successful with a tighter precision ($\gamma = \sigma = 0.001$ deg ~ 4 arc sec). Now we could anticipate that brighter stars normally result in bigger centroid error than medium bright stars, due to image blooming effects on the CCD. Also, the initial match is so critical, and there are usually ~ 20 stars in the field of view; a more conservative approach can be taken. Therefore, a conservative approach can be taken: The tighter precision tolerance ($\gamma = 3\sigma$) was used when searching for an initial candidate pivot star, whereas the looser precision threshold ($\gamma = 3\sigma$) is enforced for the each additional connected star match when the pivot method was implemented. Since the subsequent matches must simultaneously match the previously identified stars, the 3σ tolerance is still leads to a

high probability identification Figure 5 shows a better identification result using the pivot method. As shown in the figure, star 1, 6 and 14 were used as pivot stars.

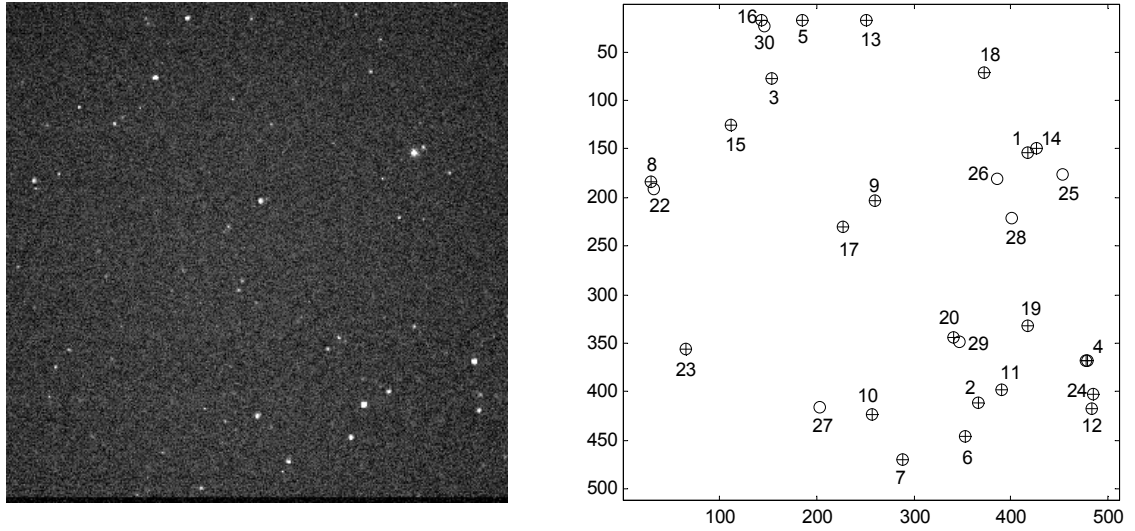
While the first part of Figure 6 shows the real night-sky image near Lyra and graphical presentation of star identification result, the second part of the figure presents the table for the identification results, including the cataloged number, sensor location, right ascension, declination, and direction cosines of identified stars. Both the pivot method and the modified SLA star identification methods were highly successful in these identifications. Further studies are underway to evaluate efficiency over a large number of night sky experiments.

Figure 5 Night-Sky Image and Star Identification Result: Taurus



#	cat#	X	Y	RA	DEC	Direction Cosines		
1	2738	192.204468	170.308151	68.9801	16.5095	0.343905	0.894972	0.284174
4	2716	380.276642	104.873596	65.7336	17.5426	0.391867	0.869247	0.301414
5	2724	301.455353	178.440033	67.1098	16.3597	0.373218	0.883953	0.281666
6	2726	299.638000	202.602936	67.1435	15.9621	0.373448	0.885952	0.275002
7	2725	296.490753	6.685185	67.154	19.1804	0.366703	0.870395	0.328544
9	2721	329.673584	278.363739	66.6519	14.7139	0.383319	0.888006	0.253992
11	2731	270.547211	188.750610	67.64	16.194	0.365331	0.888118	0.27889
12	2736	222.941879	271.536743	68.462	14.8444	0.354865	0.89913	0.256195
13	2728	298.541382	379.811249	67.2088	13.0475	0.377373	0.898122	0.225759
14	2734	263.493439	255.538467	67.7799	15.1051	0.3651	0.893753	0.260591
15	2740	159.894897	413.726105	69.5393	12.5109	0.341264	0.914665	0.216625
16	2713	417.640564	253.431641	65.1512	15.0957	0.405724	0.876106	0.260433
17	2733	271.536682	339.548248	67.6556	13.7245	0.369318	0.898507	0.237254
18	2720	332.723083	222.907700	66.5864	15.6184	0.382694	0.883777	0.269229
19	2732	269.656769	219.599258	67.6619	15.6918	0.365906	0.890485	0.270462
20	2741	144.871796	213.666672	69.7882	15.8	0.332438	0.902967	0.27228
21	2729	277.480560	222.789368	67.5356	15.6378	0.367965	0.889911	0.269556

Figure 6 Night-Sky Image and Star Identification Result: Lyra



#	cat#	X	Y	RA	DEC	Direction Cosines		
1	1890	416.871704	154.336884	-76.3741	36.8985	0.188395	-0.777192	0.600399
2	1893	366.374054	411.801910	-75.2641	32.6896	0.214075	-0.813927	0.540087
3	1904	153.192368	78.159874	-70.9079	38.1337	0.257277	-0.743306	0.617499
4	1888	478.806030	368.499268	-77.48	33.3627	0.181056	-0.815345	0.549937
5	1902	185.738876	17.257172	-71.5606	39.146	0.245304	-0.735723	0.631299
6	1894	352.893890	445.561310	-74.9968	32.1455	0.219188	-0.817836	0.532072
7	1897	288.823364	470.000000	-73.759	31.7442	0.23784	-0.816469	0.526128
8	1911	29.430668	183.519440	-68.4619	36.3179	0.295805	-0.749481	0.592265
9	1898	260.747131	203.904068	-73.1745	36.1002	0.233879	-0.773399	0.589199
10	7921	256.999329	423.763916	-73.1435	32.5016	0.244559	-0.80714	0.537323
11	94	390.847687	398.762360	-75.7436	32.9015	0.206763	-0.813748	0.543197
12	1887	483.555054	417.852600	-77.5296	32.5511	0.182014	-0.823026	0.538052
13	1899	251.643600	17.758486	-72.9449	39.155	0.227429	-0.741339	0.631421
14	1889	426.698334	149.708679	-76.5685	36.9717	0.185579	-0.77708	0.601421
15	1906	111.728462	125.683754	-70.0874	37.3307	0.270817	-0.747609	0.606415
16	1905	143.704849	17.712189	-70.6668	39.1211	0.256842	-0.732065	0.630962
17	1900	227.924942	230.763916	-72.5152	35.661	0.244112	-0.774941	0.582989
18	1892	372.800232	71.841324	-75.4921	38.2662	0.196689	-0.760106	0.619316
19	1891	416.848419	332.852081	-76.2812	33.9686	0.196685	-0.805684	0.558738
20	1895	340.736572	343.991058	-74.7699	33.802	0.218291	-0.80178	0.556325
21	1888	477.937622	368.565369	-77.48	33.3627	0.181056	-0.815345	0.549937
23	1907	65.756310	356.789154	-69.3611	33.5183	0.293864	-0.780203	0.552203
24	1886	484.669586	401.922119	-77.5589	32.8128	0.181062	-0.82071	0.541896

The basic approach pursued in both of these identification strategies is to insist that the initial triple of stars be matched with a stringent one σ precision, and form a hypothesis that these stars have been correctly matched with the corresponding cataloged stars accessed using the K-Vector method. If one or more subsequent stars have interstar angles matching all three of these stars to within 3σ , then the hypothesis of a correct match is confirmed, if not, the search is continued until the initial triple is confirmed by additional matches, or until all measured stars have been considered. We have found that

when matching 4 or more stars with this precision (< 10 arc seconds), having the correct match between measured and cataloged stars is essentially the certain (we have found no counter examples). Only for a small fraction of the measured star patterns studied ($<< 1\%$) with this conservative strategy will fail to achieve a match in the catalog. Regarding failure to match an occasional measured star pattern, our philosophy is that it is far preferable to have an occasional pattern match failure, than to incur frequent incorrect star identifications. We wish to reduce the probability of an incorrect identification to essentially zero. It is anticipated that a continuous Kalman filter will be processing the discrete attitudes determined from the identified stars, and a rare data dropout resulting from a star identification failure will not represent a significant operational concern. This is especially true in the most likely event that the spacecraft will have high quality rate gyros to measure angular velocity, which can be accurately integrated for several minutes with minimal accuracy degradation.

Plans for On-Orbit Experiments

In order to represent the on-orbit demonstration of a Lost-In-Space star identification and attitude determination process for the second prototype of the hardware, the StarNav experiment is planned for Shuttle Mission STS 107, January 2001. The second StarNav prototype is also a low cost star tracker for micro spacecraft under being developed at Texas A&M University (TAMU), which has a $7.5 \sim 8^\circ$ of single FOV with a magnitude 6.5 of sensitivity. Its camera head unit has a 70 mm f/1.0 lens, 512 x 512 CCD detector with high quantum efficiency, and thermoelectric cooler. The imager is a Pegasus-XL CCD camera which will be interfaced to the on-board computer via a parallel port that also carries the camera's power supply modules. The PC-104 embeddable single board computer has a Pentium-133 processor with 32Mb RAM and a 40Mb "solid state disk" as well as conventional drives. All of these components have been customized to significantly increase radiation hardness and ability withstand the launch and on-orbit environment. All software is being implemented in ANSI-C under the real-time environment of QNX operating system. The ground user interface support software, which is written in Microsoft Visual C++, is also being developed to send uplink commands and to acquire the downloaded images during the space mission. A minimum of ten hours of experimentation is planned, however it is hoped that 30 or 40 hours of operation will be carried out. The results of the imaging and attitude estimation results will be available in near-real time over the internet, interested parties should contact the authors to obtain final details of these plans during the fall of 2000.

Conclusions

We have presented an approach to star pattern identification and attitude determination suitable for solving the general *lost-in-space* problem wherein no prior attitude estimate is available. Results from ground test experiments are also reported which support the validity and practicality of the approach. It is planned to test these algorithms in the StarNav star tracker and processor to be flown aboard STS-107 in 2001; this will represent an important milestone toward the autonomous spacecraft envisioned for the new millennium.

References

1. Junkins, J. L., White, C. C. III, and Turner, J. D., "Star Pattern Recognition for Real Time Attitude Determination," *Journal of the Astronautical Sciences*, Vol. XXV, No.3, July-September, 1977, pp. 251-270.
2. Strikwerda, T. E. and Junkins, J. L., "Star Pattern Recognition and Spacecraft Attitude Determination," U.S. Army Engineer Topographic Laboratories, ETL-0260, Fort Belvoir, Virginia, May 1981.
3. Sasaki, T. and Kosaka, M., "A Star Identification Method for Satellite Attitude Determination using Star Sensors," *Proceedings of the 15th International Symposium on Space Technology and Sciences*, Tokyo, Japan, May 1986, pp. 1125-1130.
4. Alveda, P. and San Martin, A. M., "Neural Network Star Pattern Recognition of Spacecraft Attitude Determination and Control," *Advances in Neural Information Processing System I*, Denver, Colorado, 1988, pp. 314-322.
5. Van Bezooijen, R. W. H., "A Star Pattern Recognition Algorithm for Autonomous Attitude Determination," *Preprint of Xith IFAC Symposium on Automatic Control in Aerospace*, Tsukuba, Japan, July 1989, pp. 51-58.
6. Strikwerda, T. E., Fisher, H. L., Kilgus, C. C., and Frank, L. J., "Autonomous Star Identification and Spacecraft Attitude Determination with CCD Star Trackers," *First International Conference on Spacecraft Guidance, Navigation and Control Systems*, Noordwick, The Netherlands, June 1991, pp. 195-200.
7. Anderson, David, *Autonomous Star Sensing and Pattern Recognition for Spacecraft Attitude Determination*, Ph.D. Thesis, Texas A&M University, College Station, TX, May 1991.
8. Liebe, C. C., "Pattern Recognition of Star Constellations for Spacecraft Applications," *IEEE Aerospace Electronics Systems Magazine*, Vol. 7, 1992, pp. 34-41.
9. De Antonio, L., Udomkesmalee, S., Alexander, J. W., Blue, R., Dennison, E., Sevaston, G., and Scholl, M., "Star-tracker Based, All-sky, Autonomous Attitude Determination," *SPIE Proceedings on Space Guidance, Control, and Tracking*, Vol. 1949, 1993, pp. 204-215.
10. Udomkesmalee, S., Alexander, J. W., and Tolivar, A. F., "Stochastic Star Identification," *Journal of Guidance, Control, and Dynamics*, vol. 17, No. 6, November-December 1994, pp. 1283-1286.
11. Mortari, D., "Search-Less Algorithm for Star Pattern Recognition," *The Journal of the Astronautical Sciences*, Vol. 45, No. 2, April-June 1997, pp. 179-194.
12. Mortari, D., "SP-Search: A New Algorithm for Star Pattern Recognition," AAS Paper No. 99-181, AAS/AIAA Space Flight Mechanics Meeting, Breckenridge, CO, Feb. 7-10, 1999.
13. Mortari, D. and Angelucci, M., "Star Pattern Recognition and Mirror Assembly Misalignment for DIGISTAR II and III Multiple FOVs Star Sensors," AAS Paper No. 99-182, AAS/AIAA Space Flight Mechanics Meeting, Breckenridge, CO, Feb. 7-10, 1999.

14. Mortari, D. and Junkins J. L., "SP-Search Star Pattern Recognition for Multiple Fields of View Star Trackers," AAS Paper No. 99-437, AAS/AIAA Astrodynamics Specialist Conference, Girdwood, Alaska, Aug. 16-19, 1999.
15. Ju, G., Kim, H., Pollock, T. C. and Junkins, J. L., "Micro Star Tracker and Attitude Determination System," Third International Symposium on Reducing the Cost of Spacecraft Ground Systems and Operations, Tainan, Taiwan, Mar. 22-24, 1999.
16. Ju, G., Kim, H., Pollock, T. C. and Junkins, J. L., "DIGISTAR: A Low Cost Micro Star Tracker," AIAA Paper No. 99-4603, AIAA 1999 Space Technology Conference & Exposition, Albuquerque, NM, Sep. 28-30, 1999